

17B.2. Relations among fluxes in multicomponent systems. Verify Eqs. (K), (O), (T), and (X) of Table 17.8-1 using only the definitions of concentrations, velocities, and fluxes.

Table 17.8-1 Notation for Mass and Molar Fluxes*

Quantity	With respect to stationary axes	With respect to mass average velocity $\mathbf{v}$	With respect to molar average velocity $\mathbf{v}^*$
Velocity of species α (cm/s)	$\mathbf{v}_\alpha$ (A)	$\mathbf{v}_\alpha - \mathbf{v}$ (B)	$\mathbf{v}_\alpha - \mathbf{v}^*$ (C)
Mass flux of species α (g/cm ² s)	$\mathbf{n}_\alpha = \rho_\alpha \mathbf{v}_\alpha$ (D)	$\mathbf{j}_\alpha = \rho_\alpha (\mathbf{v}_\alpha - \mathbf{v})$ (E)	$\mathbf{j}_\alpha^* = \rho_\alpha (\mathbf{v}_\alpha - \mathbf{v}^*)$ (F)
Molar flux of species α (g-moles/cm ² s)	$\mathbf{N}_\alpha = c_\alpha \mathbf{v}_\alpha$ (G)	$\mathbf{J}_\alpha = c_\alpha (\mathbf{v}_\alpha - \mathbf{v})$ (H)	$\mathbf{J}_\alpha^* = c_\alpha (\mathbf{v}_\alpha - \mathbf{v}^*)$ (I)
Sums of mass fluxes	$\sum_{\alpha=1}^N \mathbf{n}_\alpha = \rho \mathbf{v}$ (J)	$\sum_{\alpha=1}^N \mathbf{j}_\alpha = 0$ (K)	$\sum_{\alpha=1}^N \mathbf{j}_\alpha^* = \rho (\mathbf{v} - \mathbf{v}^*)$ (L)
Sums of molar fluxes	$\sum_{\alpha=1}^N \mathbf{N}_\alpha = c \mathbf{v}^*$ (M)	$\sum_{\alpha=1}^N \mathbf{J}_\alpha = c (\mathbf{v}^* - \mathbf{v})$ (N)	$\sum_{\alpha=1}^N \mathbf{J}_\alpha^* = 0$ (O)
Relations between mass and molar fluxes	$\mathbf{n}_\alpha = M_\alpha \mathbf{N}_\alpha$ (P)	$\mathbf{j}_\alpha = M_\alpha \mathbf{J}_\alpha$ (Q)	$\mathbf{j}_\alpha^* = M_\alpha \mathbf{J}_\alpha^*$ (R)
Interrelations among mass fluxes	$\mathbf{n}_\alpha = \mathbf{j}_\alpha + \rho_\alpha \mathbf{v}$ (S)	$\mathbf{j}_\alpha = \mathbf{n}_\alpha - \omega_\alpha \sum_{\beta=1}^N \mathbf{n}_\beta$ (T)	$\mathbf{j}_\alpha^* = \mathbf{n}_\alpha - x_\alpha \sum_{\beta=1}^N \frac{M_\alpha}{M_\beta} \mathbf{n}_\beta$ (U)
Interrelations among molar fluxes	$\mathbf{N}_\alpha = \mathbf{J}_\alpha^* + c_\alpha \mathbf{v}^*$ (V)	$\mathbf{J}_\alpha = \mathbf{N}_\alpha - \omega_\alpha \sum_{\beta=1}^N \frac{M_\beta}{M_\alpha} \mathbf{N}_\beta$ (W)	$\mathbf{J}_\alpha^* = \mathbf{N}_\alpha - x_\alpha \sum_{\beta=1}^N \mathbf{N}_\beta$ (X)

* Entries in the shaded boxes, involving the "hybrid fluxes" $\mathbf{j}_\alpha^*$ and $\mathbf{J}_\alpha^*$, are seldom needed; they are included only for the sake of completeness.

for (K): $\sum_{\alpha=1}^N \mathbf{j}_\alpha = 0$ - def.: $\mathbf{j}_\alpha = \rho_\alpha (\mathbf{v}_\alpha - \mathbf{v})$

$$\sum_{\alpha} \mathbf{j}_\alpha = \sum_{\alpha} \rho_\alpha (\mathbf{v}_\alpha - \mathbf{v}) \Rightarrow \sum_{\alpha} \rho_\alpha \mathbf{v}_\alpha - \sum_{\alpha} \rho_\alpha \mathbf{v}$$

distribution

$\rightarrow$ factor terms $\Rightarrow \sum_{\alpha} \rho_\alpha \mathbf{v}_\alpha - \mathbf{v} \sum_{\alpha} \rho_\alpha$

• Total mass flux $\rightarrow \sum_{\alpha} \rho_\alpha \mathbf{v}_\alpha = \rho \mathbf{v}$
definition

• Total density $\rightarrow \sum_{\alpha} \rho_\alpha = \rho$

sub back in $\Rightarrow \rho \mathbf{v} - \mathbf{v} \rho = 0$ ✓
 $(\sum_{\alpha} \rho_\alpha \mathbf{v}_\alpha) - \mathbf{v} \sum_{\alpha} \rho_\alpha$

for (O): $\sum_{\alpha=1}^N \mathbf{J}_\alpha^* = 0$

Def: $J_{\alpha}^* = C_{\alpha}(V_{\alpha} - V^*)$

sum: $\sum_{\alpha} J_{\alpha}^* = \sum_{\alpha} C_{\alpha}(V_{\alpha} - V^*) = \sum_{\alpha} C_{\alpha} V_{\alpha} - V^* \sum_{\alpha} C_{\alpha}$

Definitions: $\left\{ \begin{array}{l} \sum_{\alpha} C_{\alpha} V_{\alpha} = C V^* \\ \sum_{\alpha} C_{\alpha} = C \\ C V^* - V^* C = 0 \end{array} \right. \checkmark$

Verify: $J_{\alpha} = n_{\alpha} - w_{\alpha} \sum_{\beta=1}^N n_{\beta}$

Def: $n_{\alpha} = \rho_{\alpha} V_{\alpha}$, $J_{\alpha} = \rho_{\alpha}(V_{\alpha} - V)$

expand J_{α} : $J_{\alpha} = \rho_{\alpha} V_{\alpha} - \rho_{\alpha} V$

Replace terms: $\circ \rho_{\alpha} V_{\alpha} = n_{\alpha}$ $\circ \rho_{\alpha} = \rho w_{\alpha}$

so $\rightarrow J_{\alpha} = n_{\alpha} - \rho w_{\alpha} V$

Replace ρV : $\sum_{\beta} n_{\beta} = \rho V$

final sub in: $J_{\alpha} = n_{\alpha} - w_{\alpha} \sum_{\beta} n_{\beta}$

Verify (x): $J_{\alpha}^* = N_{\alpha} - x_{\alpha} \sum_{\beta=1}^N N_{\beta}$

Def: $N_{\alpha} = C_{\alpha} V_{\alpha}$

$J_{\alpha}^* = C_{\alpha}(V_{\alpha} - V^*)$

expand: $J_{\alpha}^* = C_{\alpha} V_{\alpha} - C_{\alpha} V^*$

Replace terms: $\circ C_{\alpha} V_{\alpha} = N_{\alpha}$ $\circ C_{\alpha} = C x_{\alpha}$

$$\hookrightarrow J_{\alpha}^{**} = N_{\alpha} - c x_{\alpha} v^{**}$$

Replace cv^{**} : $\sum_f N_f = cv^{**}$

Final form: $J_{\alpha}^{**} = N_{\alpha} - x_{\alpha} \sum_f N_f$ ✓

17B.3. Relations between fluxes in binary systems. The following equation is helpful for interrelating expressions in mass units and those in molar units in two-component systems:

$$\frac{j_A}{\rho \omega_A \omega_B} = \frac{J_A^*}{c x_A x_B} \quad (17B.3-1)$$

Verify the correctness of this relation.

Table 17.8-1 Notation for Mass and Molar Fluxes*

Quantity	With respect to stationary axes	With respect to mass average velocity $\mathbf{v}$	With respect to molar average velocity $\mathbf{v}^*$
Velocity of species α (cm/s)	$\mathbf{v}_\alpha$ (A)	$\mathbf{v}_\alpha - \mathbf{v}$ (B)	$\mathbf{v}_\alpha - \mathbf{v}^*$ (C)
Mass flux of species α (g/cm ² s)	$\mathbf{n}_\alpha = \rho_\alpha \mathbf{v}_\alpha$ (D)	$\mathbf{j}_\alpha = \rho_\alpha (\mathbf{v}_\alpha - \mathbf{v})$ (E)	$\mathbf{j}_\alpha^* = \rho_\alpha (\mathbf{v}_\alpha - \mathbf{v}^*)$ (F)
Molar flux of species α (g-moles/cm ² s)	$\mathbf{N}_\alpha = c_\alpha \mathbf{v}_\alpha$ (G)	$\mathbf{J}_\alpha = c_\alpha (\mathbf{v}_\alpha - \mathbf{v})$ (H)	$\mathbf{J}_\alpha^* = c_\alpha (\mathbf{v}_\alpha - \mathbf{v}^*)$ (I)
Sums of mass fluxes	$\sum_{\alpha=1}^N \mathbf{n}_\alpha = \rho \mathbf{v}$ (J)	$\sum_{\alpha=1}^N \mathbf{j}_\alpha = 0$ (K)	$\sum_{\alpha=1}^N \mathbf{j}_\alpha^* = \rho (\mathbf{v} - \mathbf{v}^*)$ (L)
Sums of molar fluxes	$\sum_{\alpha=1}^N \mathbf{N}_\alpha = c \mathbf{v}^*$ (M)	$\sum_{\alpha=1}^N \mathbf{J}_\alpha = c (\mathbf{v}^* - \mathbf{v})$ (N)	$\sum_{\alpha=1}^N \mathbf{J}_\alpha^* = 0$ (O)
Relations between mass and molar fluxes	$\mathbf{n}_\alpha = M_\alpha \mathbf{N}_\alpha$ (P)	$\mathbf{j}_\alpha = M_\alpha \mathbf{J}_\alpha$ (Q)	$\mathbf{j}_\alpha^* = M_\alpha \mathbf{J}_\alpha^*$ (R)
Interrelations among mass fluxes	$\mathbf{n}_\alpha = \mathbf{j}_\alpha + \rho_\alpha \mathbf{v}$ (S)	$\mathbf{j}_\alpha = \mathbf{n}_\alpha - \omega_\alpha \sum_{\beta=1}^N \mathbf{n}_\beta$ (T)	$\mathbf{j}_\alpha^* = \mathbf{n}_\alpha - x_\alpha \sum_{\beta=1}^N \frac{M_\alpha}{M_\beta} \mathbf{n}_\beta$ (U)
Interrelations among molar fluxes	$\mathbf{N}_\alpha = \mathbf{J}_\alpha^* + c_\alpha \mathbf{v}^*$ (V)	$\mathbf{J}_\alpha = \mathbf{N}_\alpha - \omega_\alpha \sum_{\beta=1}^N \frac{M_\beta}{M_\alpha} \mathbf{N}_\beta$ (W)	$\mathbf{J}_\alpha^* = \mathbf{N}_\alpha - x_\alpha \sum_{\beta=1}^N \mathbf{N}_\beta$ (X)

* Entries in the shaded boxes, involving the "hybrid fluxes" $\mathbf{j}_\alpha^*$ and $\mathbf{J}_\alpha$, are seldom needed; they are included only for the sake of completeness.

mass flux: $\mathbf{j}_\alpha = \rho_\alpha (\mathbf{v}_\alpha - \mathbf{v})$ molar flux: $\mathbf{J}_\alpha^* = c_\alpha (\mathbf{v}_\alpha - \mathbf{v}^*)$

$\rho_\alpha = \rho x_\alpha$ $c_\alpha = c x_\alpha$

Rewrite to be in terms of velocity: $\mathbf{j}_\alpha = \rho \omega_\alpha (\mathbf{v}_\alpha - \mathbf{v})$

$\mathbf{J}_\alpha^* = c x_\alpha (\mathbf{v}_\alpha - \mathbf{v}^*)$

$\mathbf{j}_\alpha^* = M_\alpha \mathbf{J}_\alpha^* \rightarrow M_\alpha \mathbf{J}_\alpha^* = \rho_\alpha (\mathbf{v}_\alpha - \mathbf{v}^*)$

Subtract the eqns

$\mathbf{j}_\alpha - M_\alpha \mathbf{J}_\alpha^* = \rho_\alpha (\mathbf{v} - \mathbf{v}^*)$

$\mathbf{j}_\beta - M_\beta \mathbf{J}_\beta^* = \rho_\beta (\mathbf{v} - \mathbf{v}^*)$

Divide both sides by $(\mathbf{v} - \mathbf{v}^*)$ to eliminate.

$\rightarrow \frac{\mathbf{j}_\alpha - M_\alpha \mathbf{J}_\alpha^*}{\rho_\alpha} = \frac{\mathbf{j}_\beta - M_\beta \mathbf{J}_\beta^*}{\rho_\beta}$

from table Binary system identities

Sum of mass fluxes: $j_A + j_B = 0 \rightarrow j_B = -j_A$

Sum of molar fluxes: $J_A^* + J_B^* = 0 \rightarrow J_B^* = -J_A^*$

} sub into eqn above

$$\rightarrow \frac{j_A - M_B J_A^*}{\rho_A} = \frac{-j_A + M_B J_A^*}{\rho_B}$$

Bring together terms: $j_A \left(\frac{1}{\rho_A} + \frac{1}{\rho_B} \right) = J_A^* \left(\frac{M_A}{\rho_A} + \frac{M_B}{\rho_B} \right)$

$\bullet \rho_A = \rho w_A$ $\bullet \rho_B = \rho w_B$

$$\frac{1}{\rho_A} + \frac{1}{\rho_B} = \frac{1}{\rho w_A} + \frac{1}{\rho w_B} = \frac{w_A + w_B}{\rho w_A w_B} = \frac{1}{\rho w_A w_B} \quad \bullet w_A + w_B = 1$$

$\bullet \frac{M_A}{\rho_A} = \frac{1}{c_A}$ $\bullet \frac{M_B}{\rho_B} = \frac{1}{c_B}$

$$\frac{M_A}{\rho_A} + \frac{M_B}{\rho_B} = \frac{1}{c_A} + \frac{1}{c_B} = \frac{x_A + x_B}{c x_A x_B} = \frac{1}{c x_A x_B}$$

sub back in $j_A \left(\frac{1}{\rho w_A w_B} \right) = J_A^* \left(\frac{1}{c x_A x_B} \right)$

↓

$$\frac{j_A}{\rho w_A w_B} = \frac{J_A^*}{c x_A x_B}$$

what we need

17B.4. Equivalence of various forms of Fick's law for binary mixtures.

- (a) Starting with Eq. (A) of Table 17.8-2, derive Eqs. (B), (D), and (F).
 (b) Starting with Eq. (A) of Table 17.8-2, derive the following flux expressions:

$$\mathbf{j}_A = -\rho(M_A M_B / M^2) \mathcal{D}_{AB} \nabla x_A \quad (17B.4-1)$$

$$\nabla x_A = \frac{1}{c \mathcal{D}_{AB}} (x_A \mathbf{N}_B - x_B \mathbf{N}_A) \quad (17B.4-2)$$

What conclusions can be drawn from these two equations?

- (c) Show that Eq. (F) of Table 17.8-2 can be written as

$$\mathbf{v}_A - \mathbf{v}_B = -\mathcal{D}_{AB} \nabla \ln \frac{x_A}{x_B} \quad (17B.4-3)$$

Table 17.8-2 Equivalent Forms of Fick's (First) Law of Binary Diffusion

Flux	Gradient	Form of Fick's Law	
$\mathbf{j}_A$	$\nabla \omega_A$	$\mathbf{j}_A = -\rho \mathcal{D}_{AB} \nabla \omega_A$	(A)
$\mathbf{J}_A^*$	∇x_A	$\mathbf{J}_A^* = -c \mathcal{D}_{AB} \nabla x_A$	(B)
$\mathbf{n}_A$	$\nabla \omega_A$	$\mathbf{n}_A = \omega_A (\mathbf{n}_A + \mathbf{n}_B) - \rho \mathcal{D}_{AB} \nabla \omega_A = \rho_A \mathbf{v} - \rho \mathcal{D}_{AB} \nabla \omega_A$	(C)
$\mathbf{N}_A$	∇x_A	$\mathbf{N}_A = x_A (\mathbf{N}_A + \mathbf{N}_B) - c \mathcal{D}_{AB} \nabla x_A = c_A \mathbf{v}^* - c \mathcal{D}_{AB} \nabla x_A$	(D)
$\rho(\mathbf{v}_A - \mathbf{v}_B)$	$\nabla \omega_A$	$\rho(\mathbf{v}_A - \mathbf{v}_B) = -\frac{\rho \mathcal{D}_{AB}}{\omega_A \omega_B} \nabla \omega_A$	(E)
$c(\mathbf{v}_A - \mathbf{v}_B)$	∇x_A	$c(\mathbf{v}_A - \mathbf{v}_B) = -\frac{c \mathcal{D}_{AB}}{x_A x_B} \nabla x_A$	(F)

Given: $\mathbf{j}_A = -\rho \mathcal{D}_{AB} \nabla \omega_A$ ← mass based form of Fick's law
 (what we derive from)

Part A: Convert mass fraction → mole fraction

$$\frac{\mathbf{j}_A}{\rho \omega_A \omega_B} = \frac{\mathbf{J}_A^*}{c x_A x_B} \rightarrow \mathbf{j}_A = \rho \omega_A \omega_B \frac{\mathbf{J}_A^*}{c x_A x_B} \quad (\text{Eqn A})$$

Sub into Eqn (A):

$$\rho \omega_A \omega_B \frac{\mathbf{J}_A^*}{c x_A x_B} = -\rho \mathcal{D}_{AB} \nabla \omega_A$$

$$\downarrow$$

$$\omega_A \omega_B \frac{\mathbf{J}_A^*}{c x_A x_B} = -\mathcal{D}_{AB} \nabla \omega_A$$

Relation gradients:

$$\nabla \omega_A = \frac{M_A M_B}{M^2} \nabla x_A$$

$$\omega_A \omega_B \frac{\mathbf{J}_A^*}{c x_A x_B} = -\mathcal{D}_{AB} \frac{M_A M_B}{M^2} \nabla x_A$$

Simplify composition terms

$$W_A = \frac{M_A X_A}{m}, \quad W_B = \frac{M_B X_B}{m}$$

↓

$$W_A W_B = \frac{M_A M_B}{m^2} X_A X_B$$

sub in

$$\frac{M_A M_B}{m^2} X_A X_B \cdot \frac{J_A^*}{C X_A X_B} = -D_{AB} \frac{M_A M_B}{m^2} \nabla X_A$$

↓

$$J_A^* = -C D_{AB} \nabla X_A \quad (\text{Eqn B})$$

From Table : $N_A = J_A^* + C_A V^*$

sub in B : $N_A = -C D_{AB} \nabla X_A + C X_A V^*$

$$N_A = C X_A V^* - C D_{AB} \nabla X_A \quad (\text{Eqn D})$$

From Table : $J_A^* = C_A (V_A - V^*)$

$$C X_A (V_A - V^*) = -C D_{AB} \nabla X_A$$

Divide by $C X_A$

$$V_A - V^* = -\frac{D_{AB}}{X_A} \nabla X_A$$

same for B

$$V_A - V^* = \frac{D_{AB}}{X_B} \nabla X_B$$

$$C (V_A - V_B) = -\frac{C D_{AB}}{X_A X_B} \nabla X_A \quad (\text{Eqn F})$$

b) from $j_A = -\rho D_{AB} \nabla W_A$ & use $\nabla W_A = \frac{M_A M_B}{m^2} \nabla X_A$

sub: $j_A = -\rho \left(\frac{M_A M_B}{m^2} \right) D_{AB} \nabla X_A \quad \checkmark$

$$J_A^* = -C_{AB} \nabla X_A \xrightarrow[\text{both } A+B]{} J_B^* = -C_{AB} \nabla X$$

$$X_A + X_B = 1 \Rightarrow \nabla X_B = -\nabla X_A$$

↓

$$J_B^* = C_{AB} \nabla X_A$$

$$X_A N_B - X_B N_A$$

$$\bullet N_A = J_A^* + C_{AB} V^*$$

$$\text{fin result: } \nabla X_A = \frac{1}{C_{AB}} (X_A N_B - X_B N_A) \quad \checkmark$$

$$c) \text{ From Eqn (F): } C(V_A - V_B) = -\frac{C_{AB}}{X_A X_B} \nabla X_A$$

$$V_A - V_B = -\frac{D_{AB}}{X_A X_B} \nabla X_A$$

$$\text{Identity: } \Delta \ln\left(\frac{X_A}{X_B}\right) = \frac{1}{X_A} \nabla X_A - \frac{1}{X_B} \nabla X_B$$

$$\bullet \nabla X_B = -\nabla X_A$$

$$\hookrightarrow \nabla \ln\left(\frac{X_A}{X_B}\right) = \left(\frac{1}{X_A} + \frac{1}{X_B}\right) \nabla X_A = \frac{1}{X_A X_B} \nabla X_A$$

↙ substitute

$$V_A - V_B = -D_{AB} \nabla \ln\left(\frac{X_A}{X_B}\right)$$

18B.2. Error in neglecting the convection term in evaporation.

- (a) Rework the problem in the text in §18.2 by neglecting the term $x_A(N_A + N_B)$ in Eq. 18.0-1. Show that this leads to

$$N_{A2} = \frac{cD_{AB}}{z_2 - z_1} (x_{A1} - x_{A2}) \quad (18B.2-1)$$

This is a useful approximation if A is present only in very low concentrations.

- (b) Obtain the result in (a) from Eq. 18.2-14 by making the appropriate approximation.
 (c) What error is made in the determination of D_{AB} in Example 18.2-2 if the result in (a) is used?

Part a) $x_A (N_A + N_B) \approx 0 \rightarrow N_A = -cD_{AB} \frac{dx_A}{dz}$

Rearrange to solve for dx_A : $\frac{dz_A}{dz} = -\frac{N_A}{cD_{AB}}$

$$dx_A = -\frac{N_A}{cD_{AB}} dz$$

Integrate: $z_1 \rightarrow z_2$
 $x_{A1} \rightarrow x_{A2}$

$$\int_{x_{A1}}^{x_{A2}} dx_A = -\frac{N_A}{cD_{AB}} \int_{z_1}^{z_2} dz$$

$$x_{A2} - x_{A1} = -\frac{N_A}{cD_{AB}} (z_2 - z_1)$$

• solve for N_A

$$x_{A2} - x_{A1} = -\frac{N_A}{cD_{AB}} (z_2 - z_1)$$

$$\rightarrow N_A = \frac{cD_{AB}}{z_2 - z_1} (x_{A1} - x_{A2})$$

Part b) $N_A = \frac{cD_{AB}}{z_2 - z_1} \ln \left(\frac{1 - x_{A2}}{1 - x_{A1}} \right)$

Apply Approximation (dilute A)

• if $x_A \ll 1$ then $\rightarrow 1 - x_A \approx 1$

taylor series $\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \dots - \frac{x^n}{n}$ • $x \rightarrow 0$

by taylor expansion $\ln(1-x_n) \approx -x_n$

$$\ln\left(\frac{1-x_{n2}}{1-x_{n1}}\right) = \ln(1-x_{n2}) - \ln(1-x_{n1})$$

$$\approx (-x_{n2}) - (-x_{n1}) = x_{n1} - x_{n2}$$

sub back in:

$$N_A = \frac{C D_{AB}}{z_2 - z_1} (x_{n1} - x_{n2})$$

same as part 1

18B.5. Absorption of chlorine by cyclohexene. Chlorine can be absorbed from Cl_2 -air mixtures by olefins dissolved in CCl_4 . It was found⁵ that the reaction of Cl_2 with cyclohexene (C_6H_{10}) is second order with respect to Cl_2 and zero order with respect to C_6H_{10} . Hence the rate of disappearance of Cl_2 per unit volume is $k_2'' c_A^2$ (where A designates Cl_2).

Rework the problem of §18.4 where B is a C_6H_{10} - CCl_4 mixture, assuming that the diffusion can be treated as pseudobinary. Assume that the air is essentially insoluble in the C_6H_{10} - CCl_4 mixture. Let the liquid phase be sufficiently deep that L can be taken to be infinite.

(a) Show that the concentration profile is given by

$$\frac{c_{A0}}{c_A} = \left[1 + \sqrt{\frac{k_2'' c_{A0}}{6D_{AB}}} z \right]^2 \quad (18B.5-1)$$

(b) Obtain an expression for the rate of absorption of Cl_2 by the liquid.

(c) Suppose that a substance A dissolves in and reacts with substance B so that the rate of disappearance of A per unit volume is some arbitrary function of the concentration, $f(c_A)$. Show that the rate of absorption of A is given by

$$N_{Az}|_{z=0} = \sqrt{2D_{AB} \int_0^{c_{A0}} f(c_A) dc_A} \quad (18B.5-2)$$

Use this result to check the result of (b).

Diffusion of $A \rightarrow$ liquid

Rxn $\rightarrow -r_A = k_2''' c_A^2$ this is 2nd order

shell bal! $\frac{d}{dz} \left(-D_{AB} \frac{dc_A}{dz} \right) = -r_A$

$D_{AB} \frac{d^2 c_A}{dz^2} = k_2''' c_A^2 \rightarrow \bullet$ multiply both sides by $\frac{dc_A}{dz}$

$D_{AB} \frac{d^2 c_A}{dz^2} \frac{dc_A}{dz} = k_2''' c_A^2 \frac{dc_A}{dz}$

$\frac{d^2 c_A}{dz^2} \frac{dc_A}{dz} = \frac{1}{2} \frac{d}{dz} \left(\frac{dc_A}{dz} \right)^2 \quad c_A^2 \frac{dc_A}{dz} = \frac{1}{3} \frac{d}{dz} (c_A^3)$

• rewrite

$D_{AB} \frac{1}{2} \frac{d}{dz} \left(\frac{dc_A}{dz} \right)^2 = k_2''' \cdot \frac{1}{3} \frac{d}{dz} (c_A^3)$

1st integration $\rightarrow \frac{D_{AB}}{2} \left(\frac{dc_A}{dz} \right)^2 = \frac{k_2'''}{3} c_A^3 + C$

BC's $\begin{cases} c_A(z=0) = c_{A0} \\ z \rightarrow \infty, c_A = 0 \\ z \rightarrow \infty, \frac{dc_A}{dz} = 0 \end{cases}$

so $C = 0$

$\frac{D_{AB}}{2} \left(\frac{dc_A}{dz} \right)^2 = \frac{k_2'''}{3} c_A^3$

solve gradient: $\left(\frac{dC_A}{dz}\right)^2 = \frac{2K_2''''}{3D_{AB}} C_A^3$

$$\frac{dC_A}{dz} = -\sqrt{\frac{2K_2'''}{3D_{AB}}} C_A^{3/2} \quad \text{neg b/c } C \downarrow \text{ with } z$$

integrate $\int \frac{dC_A}{C_A^{3/2}} = -\sqrt{\frac{2K_2'''}{3D_{AB}}} dz$

$$\int C_A^{-3/2} dC_A = -2C_A^{-1/2} = -\sqrt{\frac{2K_2'''}{3D_{AB}}} z + C$$

$$\downarrow$$

$$-2C_{A0}^{-1/2} = C$$

• Apply BC 1
@ $z=0$, $C_A = C_{A0}$

$$\left[\frac{1}{\sqrt{C_A}} \right] = \left[\frac{1}{\sqrt{C_{A0}}} + \sqrt{\frac{K_2'''}{6D_{AB}}} z \right]^2$$

$$\frac{C_{A0}}{C_A} = \left[1 + \sqrt{\frac{K_2''' C_{A0}}{6D_{AB}}} z \right]^2$$

Part (b): flux at surface $N_A = -D_{AB} \left(\frac{dC_A}{dz} \right)_{z=0}$

Sub in $\frac{dC_A}{dz} = -\sqrt{\frac{2K_2'''}{3D_{AB}}} C_A^{3/2}$

• BC $C_A(z=0) = C_{A0}$

$$N_A = D_{AB} \sqrt{\frac{2K_2'''}{3D_{AB}}} C_{A0}^{3/2}$$

$$N_A = \sqrt{\frac{2D_{AB}K_2'''}{3}} C_{A0}^{3/2}$$

Part (c) $N_A = \sqrt{2D_{AB} \int_0^{C_{A0}} f(C_A) dC_A}$

• $f(C_A) = k_2''' C_A^2$ Sub in

$$\int_0^{C_{A0}} k_2''' C_A^2 dC_A = k_2''' \frac{C_{A0}^3}{3}$$

plug into formula via

$$N_A = \sqrt{2D_{AB} \cdot \frac{k_2''' C_{A0}^3}{3}}$$

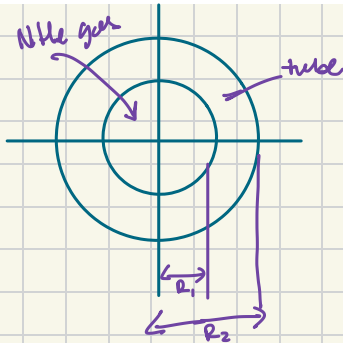
↓

$$N_A = \sqrt{\frac{2D_{AB} k_2'''}{3}} C_{A0}^{3/2} \quad \text{part b is verified}$$

18B.8. Method for separating helium from natural gas (Fig. 18B.8). Pyrex glass is almost impermeable to all gases but helium. For example, the diffusivity of He through pyrex is about 25 times the diffusivity of H_2 through pyrex, hydrogen being the closest "competitor" in the diffusion process. This fact suggests that a method for separating helium from natural gas could be based on the relative diffusion rates through pyrex.⁷

Suppose a natural gas mixture is contained in a pyrex tube with dimensions shown in the figure. Obtain an expression for the rate at which helium will "leak" out of the tube in terms of the diffusivity of helium through pyrex, the interfacial concentrations of helium in the pyrex, and the dimensions of the tube.

Answer: $W_{He} = 2\pi L \frac{D_{He-Pyrex}(c_{He,1} - c_{He,2})}{\ln(R_2/R_1)}$



- Radial diffusion
- $A = 2\pi rL$

Shell Bal: $\underbrace{(2\pi rL N_{Ar})_r}_{2\pi L} - \underbrace{(2\pi rL N_{Ar})_{r+\Delta r}}_{2\pi L} \stackrel{ss}{=} 0$
 + take limit $r \rightarrow \infty$

$$\int \frac{d}{dr} (r N_{Ar}) = 0$$

$$= N_{Ar} = C/r$$

• Use Fick's law

$$N_{Ar} = -D_{AB} \frac{dc_A}{dr} \rightarrow -D_{AB} \frac{dc_A}{dr} = \frac{C}{r}$$

$$\int dc_A = -\frac{C}{D_{AB}} \int \frac{1}{r} dr \Rightarrow \frac{dc_A}{dr} = -\frac{C}{D_{AB}} \cdot \frac{1}{r}$$

$$c_A = -\frac{C}{D_{AB}} \ln r + C_2$$

$$BCs \begin{cases} c_A(r=R_1) = c_{A1} \\ c_A(r=R_2) = c_{A2} \end{cases}$$

$$C_{A1} = -\frac{C}{D_{AB}} \ln R_1 + C_2$$

$$C_{A2} = -\frac{C}{D_{AB}} \ln R_2 + C_2$$

$$C_{A1} - C_{A2} = -\frac{C}{D_{AB}} (\ln R_1 - \ln R_2)$$

> simplify

$$\ln R_1 - \ln R_2 = \ln(R_1/R_2) \rightarrow C_{A1} - C_{A2} = -\frac{C}{D_{AB}} \ln(R_1/R_2)$$

↓

$$C_{A1} - C_{A2} = \frac{C}{D_{AB}} \ln(R_2/R_1) \leftarrow \{-\ln(R_1/R_2)\}$$

$$C = \frac{D_{AB}(C_{A1} - C_{A2})}{\ln(R_2/R_1)}$$

$$\bullet N_{Ar} = \frac{C}{r}$$

↓

$$N_{Ar} = \frac{D_{AB}(C_{A1} - C_{A2})}{2 \ln(R_2/R_1)}$$

now we want to convert our flux into total rate

$$W_A = (A)(N_A) \quad \bullet A = 2\pi rL$$

$$W_A = (2\pi rL) \left(\frac{D_{AB}(C_{A1} - C_{A2})}{r \ln(R_2/R_1)} \right)$$

$$W_{He} = \frac{2\pi L D_{He-Ar,ex} (C_{He1} - C_{He2})}{\ln(R_2/R_1)}$$